Importance of N2Hx Reaction Pathways in Modelling of NH3/H2/Air Flame Chemistry

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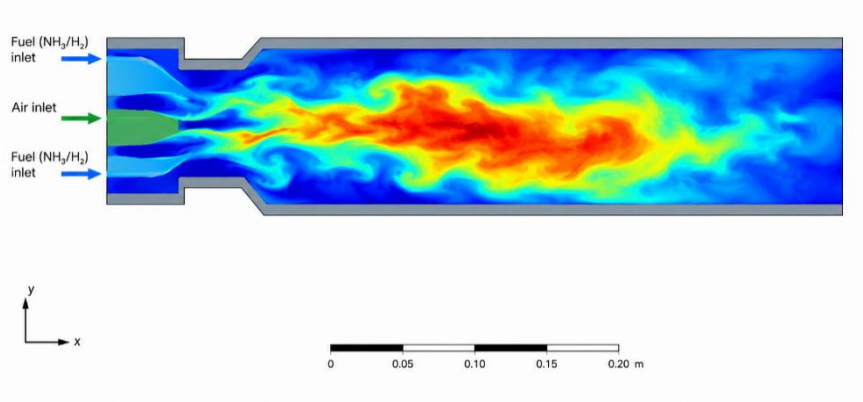
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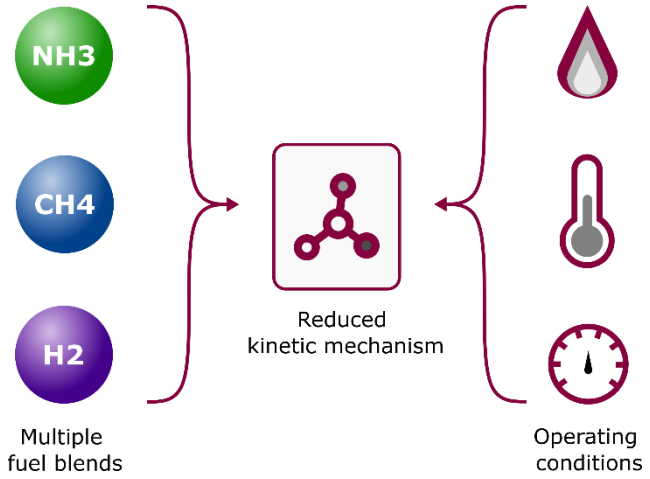


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Background

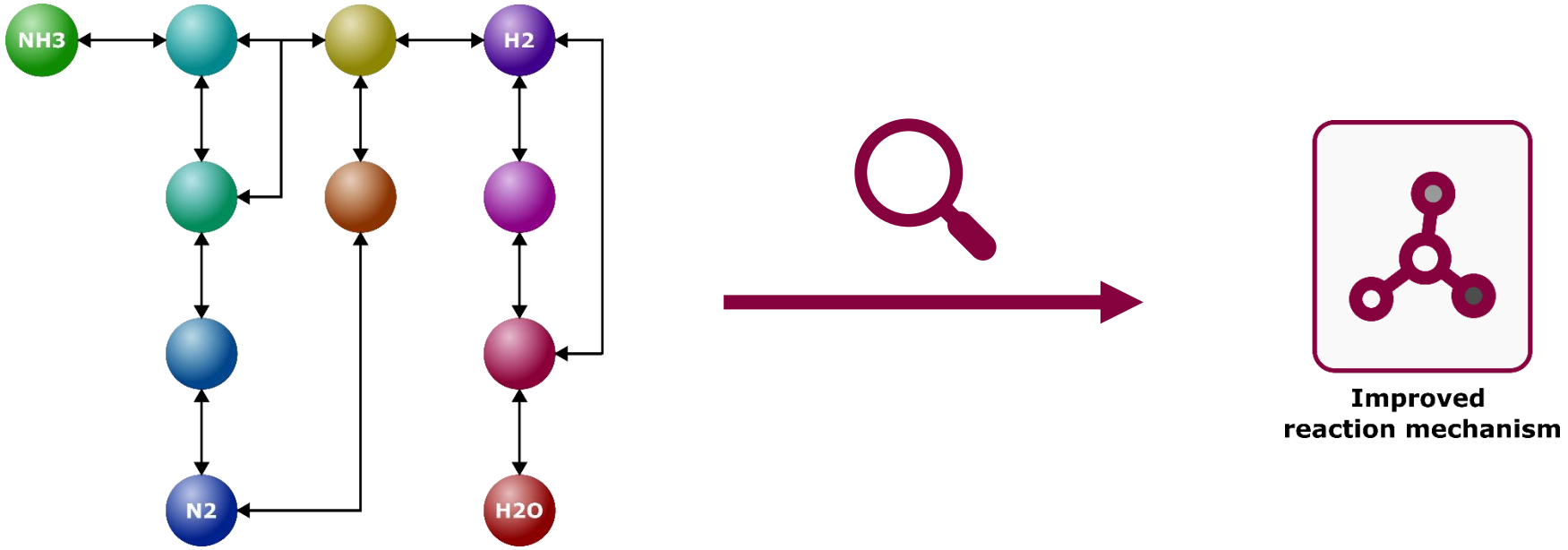
- Numerically simulating ammonia blends is necessary for combustion system development.
- Large-scale, 3D modelling with detailed chemical kinetics mechanisms increases computational cost.
- Reduced chemical kinetic mechanisms optimized for multiple ammonia blends and operating conditions is needed.





Background

- Existing reduced chemical kinetics mechanisms target specific fuel blends for select operating conditions.
- Elucidating reaction pathways particular to specific fuel blends helped develop a kinetics mechanism viable for a wider range of fuels.



Objectives

- Develop a reduced kinetic mechanism optimized for NH3, NH3/CH4 and NH3 H2
  - Prioritizing unstretched laminar burning velocity predictions
  - Viable for cases with varying pressure and equivalence ratio
- Validate the Present Mechanism with experimental flame speed measurements from literature

Development of the Present Mech

- Present mechanism is based on the reduced mechanism developed by Okafor et al. (2019).
  - This mechanism already satisfactorily models laminar burning velocity of NH3/air and NH3/CH4/air flame.
  - Fails to model NH3/H2/air flames at hydrogen concentrations greater than $X_{\mathrm{H2}} = 0.3$
- Reaction mechanism by Gotama et al. (2022) was used to study the differences in ammonia and hydrogen kinetics interaction.
  - This mechanism was optimized for laminar burning velocity of NH3/H2/air flames.

Numerical Procedure

- Unstretched laminar burning velocity, S_L, of premixed flames calculated with 1D unstretched adiabatic simulations
  - Premixed Laminar Flame Speed Calculation model in ANSYS Chemkin-PRO used
  - Computational domain of 10 cm
  - Adaptive grid settings GRAD and CURV set to 0.01

- X_{H2}, the ratio of H_2 in the fuel, was varied
  - Pressure was kept constant at 1 atm
  - Fuel-oxidizer ratio was maintained at stoichiometric equivalence ratio

Development of the Present Mech

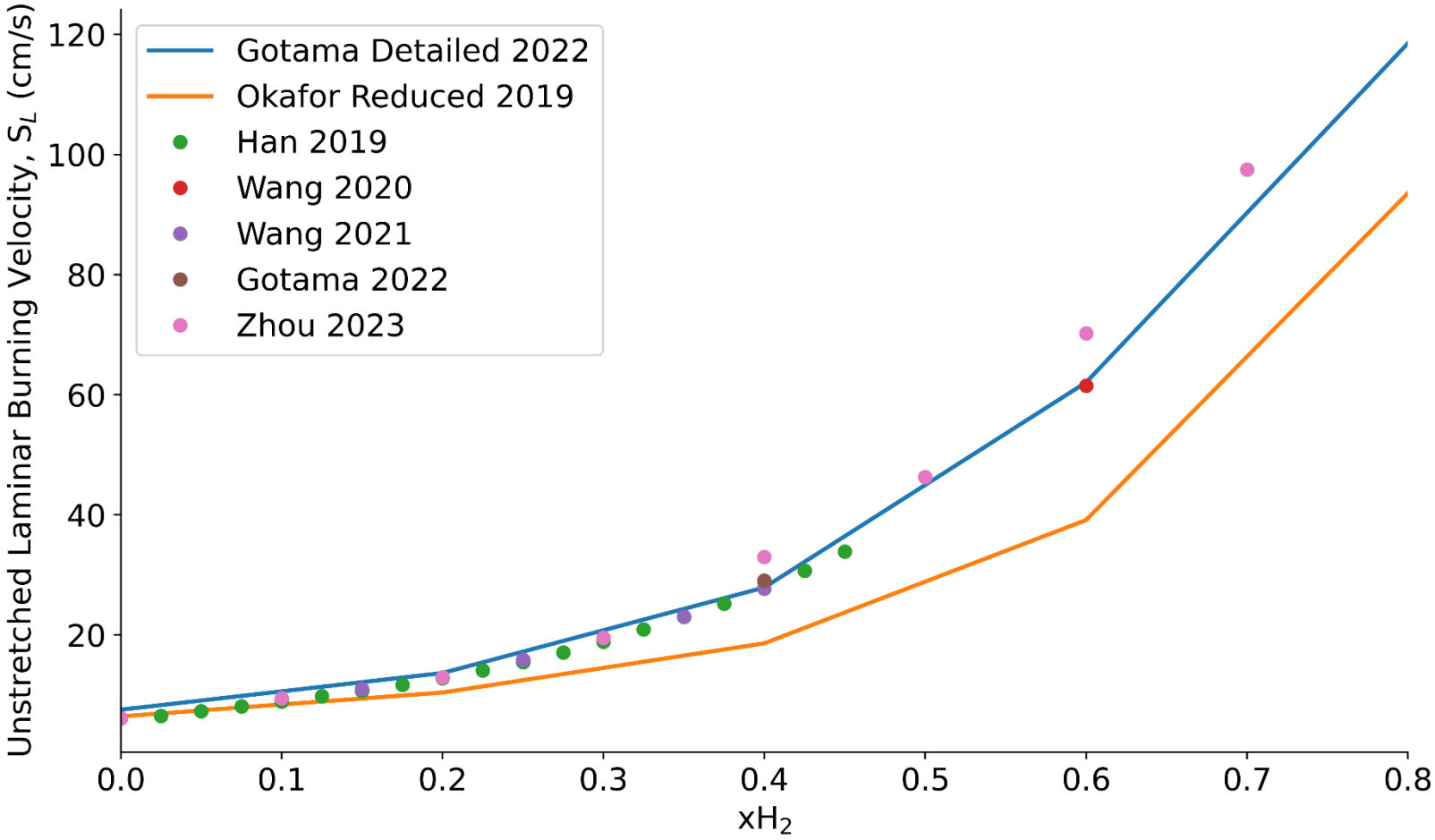


Figure: Experimental and numerical laminar burning velocity of NH3/H2/air flames under stoichiometric condition at various hydrogen concentrations.

Development of the Present Mech

- Mechanisms were compared using reaction pathways constructed with integrated rate of production (IROP) of major NH3 and H2 oxidation species.
- IROP was calculated with the equation:

$$IROP = \int_0^L \dot{\omega}_{R,i}$$

- Flame used for the reaction pathways had hydrogen concentration of $X_{\mathrm{H2}} = 0.5$ and equivalence ratio of 1.0 under atmospheric conditions.

Reaction pathways of NH3/H2/air flame at $X_{H2} = 0.5$ and $\phi = 1.0$

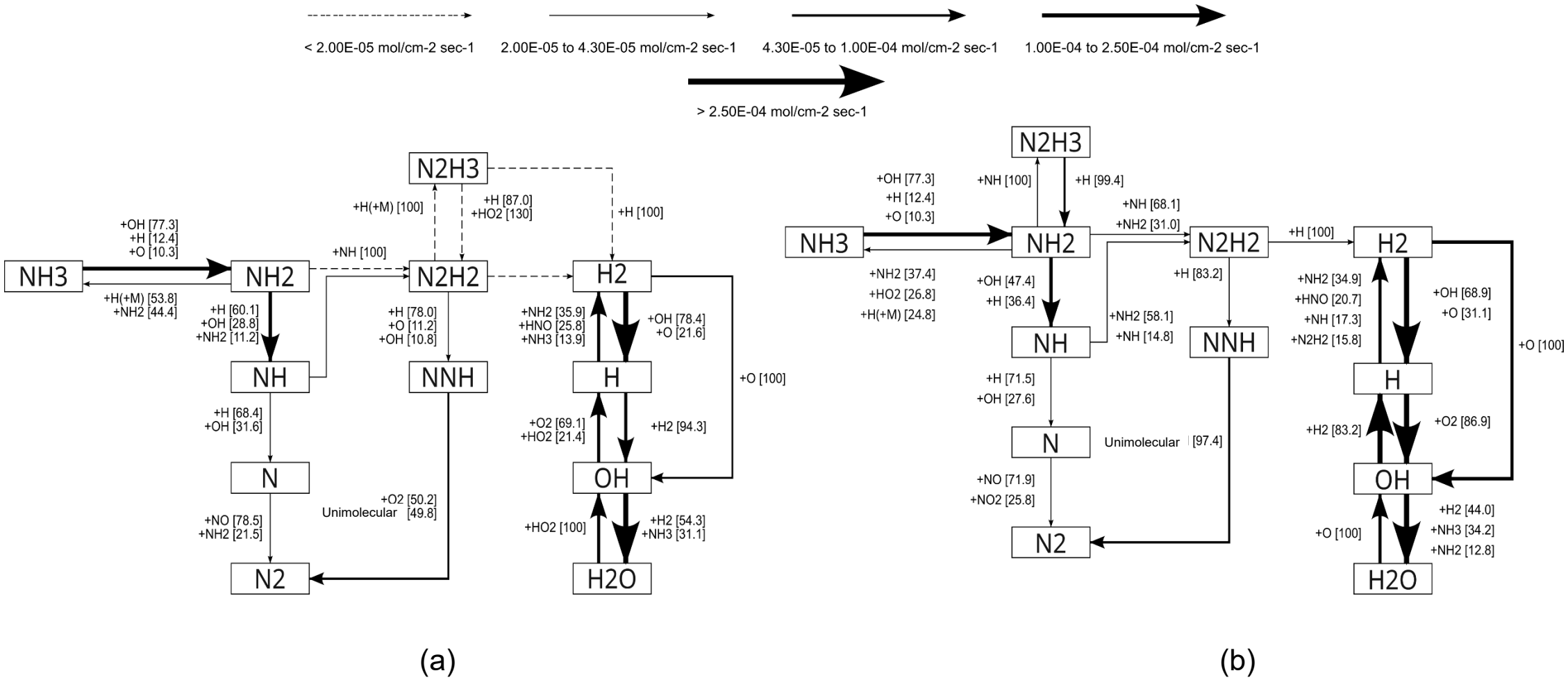


Figure: Reaction pathway of NH3/H2/air flame at $X_{H2} = 0.5$ and equivalence ratio of 1.0 using (a) Okafor Reduced Mech and (b) Gotama Mech.

Key Differences in Reaction Pathways

- $NH_2 \leftrightarrow N_2H_3$ reaction pathway was not modelled in Okafor Reduced Mech.
  - Formation of N_2H_3: $NH_2 + NH = N_2H_3$
  - Consumption of N_2H_3: $N_2H_3 + H = 2NH_2$
- $NNH \rightarrow N_2$ pathway in Gotama Mech was dominated by $NNH = N_2 + H$
  - Contributes to H radical production
- IROP of $NH_2 \rightarrow N_2H_2$ pathway was higher in Gotama Mech.
  - $NH_2 + NH = N_2H_2 + H$ contributes the most to this pathway.
- N_2H_2 formation further promotes $N_2H_2 \rightarrow NNH \rightarrow N_2$ pathways.

Present Mech

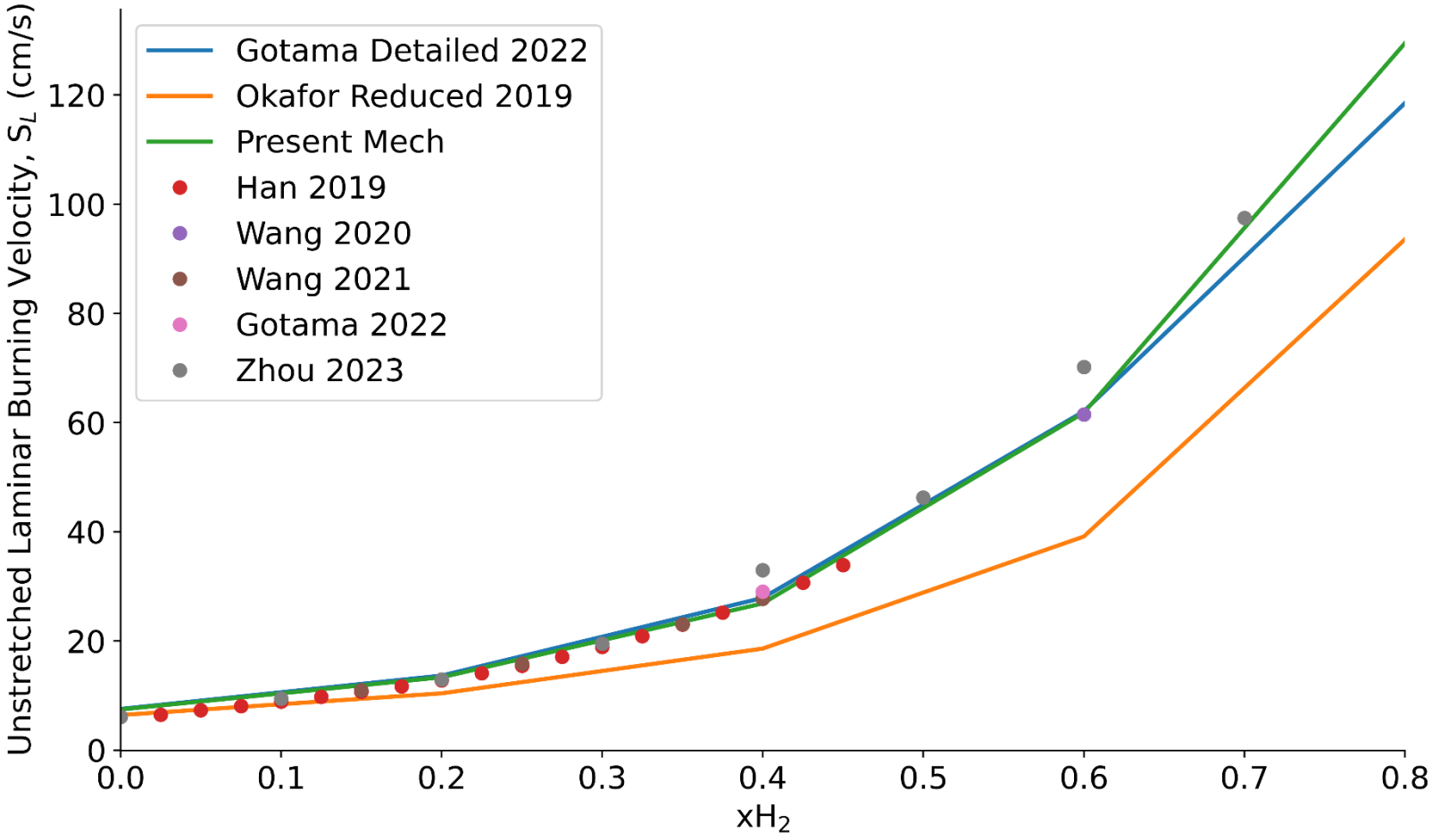


Figure: Experimental and numerical laminar burning velocity of NH3/H2/air flames under stoichiometric condition at various hydrogen concentrations.

Validation – NH3/H2/air

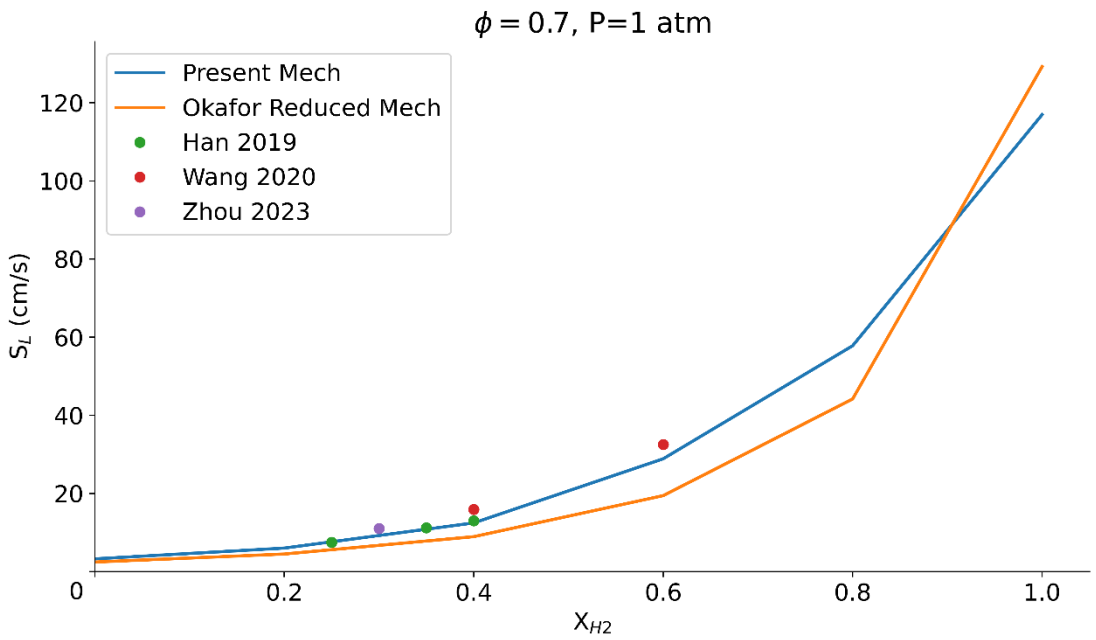


Figure 1: Comparison of measured and simulated laminar burning velocity of NH3/H2/air flames under lean condition ($\phi = 0.7$) and 1 atm of pressure.

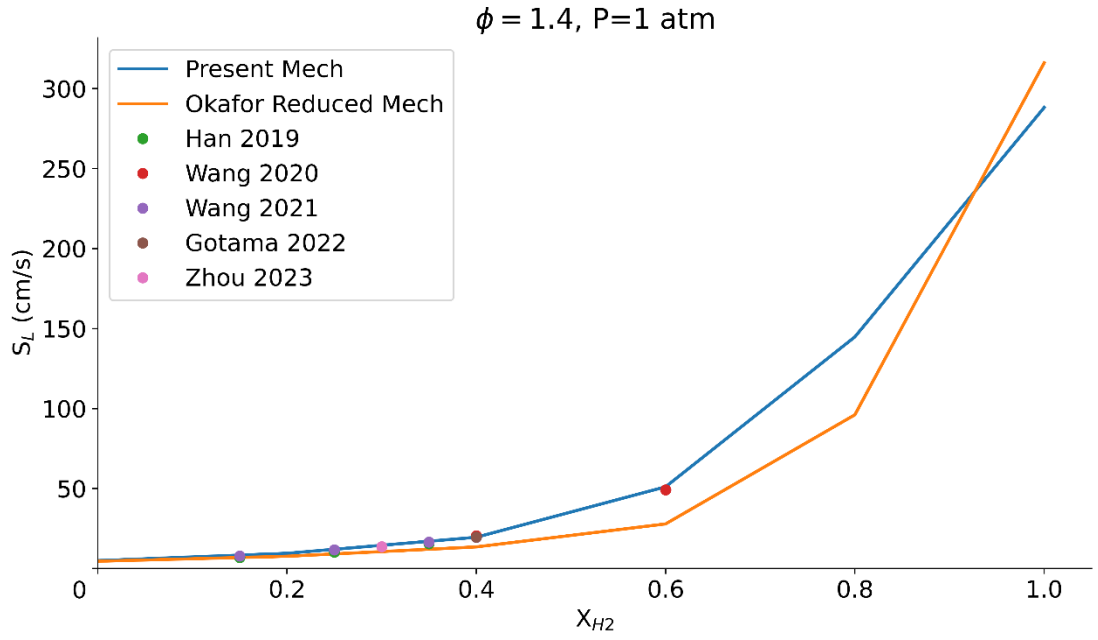


Figure 2: Comparison of measured and simulated laminar burning velocity of NH3/H2/air flames under rich condition ($\phi = 1.4$) and 1 atm of pressure.

Validation – NH3/CH4/air

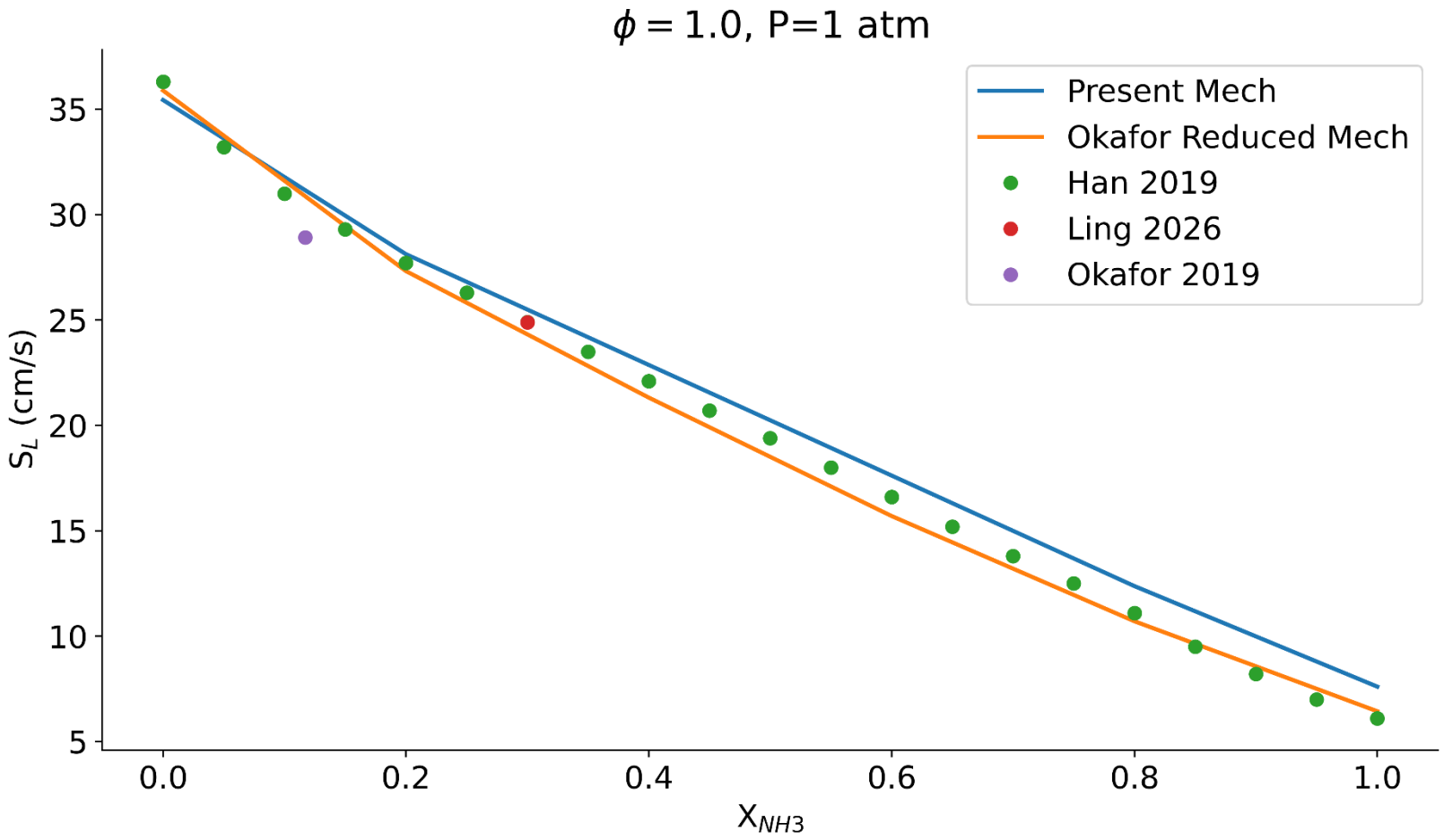


Figure: Comparison of measured and simulated laminar burning velocity of NH3/CH4/air flames under stoichiometric condition and 1 atm of pressure.